



**BRAC University**

Department of Mathematics and Natural Sciences

LECTURE ON

**Complex Variables & Laplace Transform (MAT215)**

# Laplace Transform of Step Functions

## The Second Translation Formula

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CONDUCTED BY

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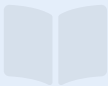
## What is Step Function?

### The Unit Step Function

The unit step function, denoted by  $u(t)$ , is defined as 

$$u(t) = \begin{cases} 0, & t < 0, \\ 1, & t \geq 0. \end{cases}$$

### Shifted Unit Step Function

The unit step function, denoted by  $u(t - a)$ , is defined as 

$$u(t - a) = \begin{cases} 0, & t < a, \\ 1, & t \geq a. \end{cases}$$

## Physical Interpretation of Unit Step Function

## The second translation formula

### Laplace Transform of $f(t) \cdot u(t - a)$

If the Laplace transform of  $f(t)$  exists, then the Laplace transform of  $f(t) \cdot u(t - a)$  is given by

$$\mathcal{L}\{f(t) \cdot u(t - a)\} = e^{-as} \mathcal{L}\{f(t + a)\},$$

where  $a > 0$ .

## Important Trigonometric Identities

### Compound Angle Identities

$$\sin(A + B) = \sin A \cos B + \cos A \sin B,$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B,$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B.$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B.$$

### Product-to-Sum Identities

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B),$$

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B),$$

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B).$$

### Negative Angle Identities

$$\sin(-A) = -\sin(A),$$

$$\cos(-A) = \cos(A).$$

How to compute  $f(t + a)$ ?

## Application of the second translation formula

### ② Problem

Evaluate

$$\mathcal{L}\{(2t - 3) u(t - 1)\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{e^{-2t} u(t - 1)\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{t^2 e^{2t} u(t - 2)\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{t \sin(2t) u(t - \pi)\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{e^{-2t} \cos t \, u(t - 3\pi)\}$$



🔍 Problem

Evaluate

$$\mathcal{L}\{te^{3t} \sin t \, u(t - \pi)\}$$



## Converting a piecewise function into Step function

## Laplace Transform of Piecewise Functions

### ② Problem

Find the Laplace transform of the function

$$f(t) = \begin{cases} 0, & 0 \leq t < 2, \\ 3, & 2 \leq t < 4, \\ e^t, & t \geq 4. \end{cases}$$

### Problem

Find the Laplace transform of the function



$$f(t) = \begin{cases} 0, & 0 \leq t < \pi, \\ \cos t, & t \geq \pi. \end{cases}$$

### Problem

Find the Laplace transform of the function



$$f(t) = \begin{cases} 5 \sin t, & 0 \leq t < \pi, \\ 0, & t \geq \pi. \end{cases}$$

### Problem

Find the Laplace transform of the function

$$f(t) = \begin{cases} 0, & 0 \leq t < \pi, \\ \sin t, & \pi \leq t < 2\pi, \\ 0, & t \geq 2\pi. \end{cases}$$



### Problem

Find the Laplace transform of the function



$$f(t) = \begin{cases} 5 \sin t, & 0 \leq t < \pi, \\ -4 \cos(2t), & t \geq \pi. \end{cases}$$

# Thank You!

We'd love your questions and feedback.

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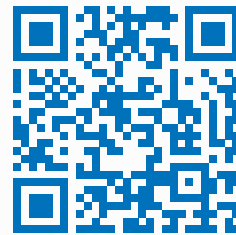
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(Lectures, walkthroughs, and course updates)



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## References

- [1] Murray R. Spiegel, *Schaum's Outline of Complex Variables*, 2nd Edition, McGraw-Hill, 2009.
- [2] Dennis G. Zill, *A First Course in Differential Equations with Modeling Applications*, 11th Edition, Cengage, 2018.